

Survey Microdata Intelligence & Quality Audit Report

Automated statistical quality assessment, multi-factor anomaly diagnostics, and supervisory compliance report for active survey dataset **chhv12025.csv**.

Dataset File: chhv12025.csv	Dataset ID: 8083ec2b	Audit Date: 16-Aug-2026 12:03 UTC
Total Sample: 270,472 Records	Total Variables: 48 Attributes	Audit Status: COMPLETED (PASS)

1. Executive Summary & Quality Scorecards

TOTAL SURVEY RECORDS 270,472 100% Evaluated by AI Engine	DATA COMPLETENESS RATE 98.12% 1.88% Missing Attribute Rate	HIGH-RISK FLAGGED RECORDS 375 0.14% Priority Review Index	ISOLATION FOREST ANOMALIES 26,070 9.64% Multivariate Outliers
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Audit Findings Synthesis: The survey microdata dataset (chhv12025.csv) underwent comprehensive multi-tier data verification across 270,472 household observations, representing an extrapolated national household population of **3,244,878**. The dataset demonstrates an overall data integrity completeness score of **98.12%**. A total of **3,437 deterministic rule discrepancies** and **26,070 multi-dimensional Isolation Forest outliers** were flagged for supervisory investigation. Overall risk assessment categorizes **244,186 records (90.3%)** as normal, **25,911 records** as medium caution, and **375 records** as high-priority audit items.

2. Microdata Statistical Profiling & Key Measures

Variable	Mean	Median	Min – Max	IQR Range (Q1 - Q3)	Skew	Outliers (1.5x IQR)
month	6.50	6.00	1.0 – 12.0	3.0 – 10.0	0.00	0 (0%)
hh_size	4.25	4.00	1.0 – 30.0	3.0 – 5.0	1.04	8,358 (3.09%)
hce1	11,304.48	9,500.00	0.0 – 320,500.0	6,400.0 – 14,600.0	2.94	10,677 (3.95%)
hce2	803.08	0.00	0.0 – 30,000.0	0.0 – 1,200.0	2.91	17,927 (6.63%)
hce3	496.30	400.00	0.0 – 200,000.0	100.0 – 700.0	100.92	8,613 (3.18%)
hce4	11,921.64	10,000.00	0.0 – 350,000.0	6,000.0 – 15,000.0	4.04	11,982 (4.43%)
hce5	9,424.34	4,200.00	0.0 – 3,650,000.0	1,500.0 – 11,200.0	33.90	17,235 (6.37%)
hce_tot	14,382.71	12,300.00	400.0 – 332,333.0	8,692.0 – 17,667.0	3.41	12,374 (4.57%)

3. Socioeconomic Stratification & Sectoral Comparison

Sector Category	Sample Records	Sample Share (%)	Mean Expenditure (₹)	Median Income (₹)	Avg Household Size
Rural	148,718	54.98%	₹12,460.53	₹1,250.00	4.41 persons
Urban	121,754	45.02%	₹16,730.57	₹1,200.00	4.04 persons

4. Geographic & Regional Distribution Highlights

State / Region	Sample Records	National Share	Mean Expenditure (₹)	Avg Household Size
State #9	36,774	13.60%	₹11,527.35	4.79 persons
State #27	19,527	7.22%	₹14,544.61	3.97 persons
State #23	19,429	7.18%	₹13,770.29	4.68 persons
State #10	17,952	6.64%	₹11,237.31	4.76 persons
State #8	17,350	6.41%	₹15,301.40	4.66 persons
State #33	14,214	5.26%	₹15,346.71	3.48 persons

5. AI Quality Diagnostics & Priority Flagged Records

Records flagged below combine deterministic business rules (35%), Isolation Forest unsupervised ML anomalies (35%), enumerator skew (15%), and spatial cluster variance (15%):

Record #	Risk Score	Level	Rule Violation	ML Anomaly	Primary Deviation Detail
#40	70/100	Medium	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded
#45	70/100	Medium	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded
#60	70/100	Medium	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded
#70	70/100	Medium	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded
#118	70/100	Medium	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded
#142	70/100	Medium	VIOLATION	OUTLIER	Multi-feature outlier & range threshold exceeded

6. Supervisory Recommendations & Compliance Directives

- **Priority Triage:** 375 high-risk observations require formal supervisor review via the validation queue prior to national statistical aggregation.
- **Integrity Corrections:** 3,437 rule violations involve field boundary checks (negative income, invalid codes, or missing values) and should be reconciled.
- **Statistical Outliers:** 26,070 multivariate Isolation Forest anomalies require analytical spot-checking for abnormal expenditure-to-income ratios.

CERTIFIED BY DATA SUPERVISOR	ENDORSED BY DIRECTOR / HEAD OF OPERATIONS
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